Exploring How True Information Can Rival Disinformation in Going Viral – Milestone 4

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Falsehoods often seem to spread “farther, faster, deeper, and more broadly” on social media than factual content (Vosoughi, Roy, & Aral, 2018). Sensational disinformation can capture attention and disseminate across the online town square remarkably fast, frequently extinguishing any attempt at a response. This analysis aims to identify strategies and mechanisms to help accurate, true information achieve virality comparable to or greater than that of disinformation. In other words, how can we make the truth as compelling and widely shared as the lies?

Research has demonstrated that false news spreads faster than true news on social platforms like Twitter. A study by Vosoughi et al. (2018) found that false news was 70% more likely to be retweeted than true news and reached 1,500 people six times faster on average. This indicates that lies have a built-in advantage in terms of speed and reach.

There has been plenty of research exploring how to stem the flow of disinformation. MIT Sloan published a study showing that asking social media users to consider the accuracy of the news they are about to share leads to improved content quality (MIT Sloan Office of Communications, 2021). A later study by Altay et al. (2025) found that “prompting social media users to follow news organizations increases… awareness of true news stories.” And recently, Slaughter et al. (2025) noted that “crowd-sourced fact-checking” via the Community Notes feature on X “reduces the depth and structural virality” of all noted posts, regardless of if true or false.

However, I believe that simply diminishing its distribution is not enough. Stopping false information from spreading is only half the battle: we must also overwhelm the town square with true information. False stories enjoy virality because humans are habitually

driven to share sensational or provocative claims that trigger strong emotions (Ceylan, Anderson, & Wood, 2023). If our aim is to preserve and disseminate truth, then we need strategies that drive humans to share true stories with at least the same earnestness as false stories.

Metrics Of a Successful Study

A successful study will present the general elements of factual posts that either successfully or unsuccessfully regularly attain high engagement metrics (views, shares, likes, and comments) on par with or exceeding those of trending false narratives. Ideally, we would develop a formula that would reliably increase engagement metrics on truthful posts, but evidence of practices that do not work would be useful as well. Our hope is to identify statistically significant social media practices that consistently see true information being reshared by users at rates similar to popular disinformation and traveling as fast through networks.

If we can crack the code to boost truth virality, the benefits would be widespread. Social media users and society at large would receive more reliable information, reducing harm from rumors and lies. Public agencies, health experts, and educators would be better able to broadcast life-saving facts at scale. Journalists and fact-checkers would gain confidence that their hard work clarifying facts won't be drowned out by falsehoods. Businesses and brands stand to gain, as their communications and reputations would be less likely to be undermined by viral disinformation campaigns. Social media platform companies themselves would enjoy greater trust and user satisfaction if truthful content thrives, potentially easing regulatory pressures. By understanding the dynamics of

information spread and focusing on making truth as engaging and shareable as lies, we can create a more informed and resilient society.

Defining Data Requirements

To answer the question, "How can true information rival disinformation in going viral?", we need to analyze the spread and engagement of both true and false content on social media.

Needed Variables

To effectively analyze the spread of true versus false information on social media, several key data features must be considered.

Posts can be categorized by content type, labeling posts as true, false, corrections, debunks, opinions, or satire. These posts can be further categorized by visual elements, such as images, memes, videos, and infographics. Sentiment scores, such as anger, awe, surprise, sadness, trust, or disgust, capture the emotional response elicited by a post, while novelty scores indicate whether the content presents new or unexpected information (Trunfio & Rossi, 2021). These classifications provide a foundation for understanding nature and intent behind each post.

Engagement metrics help measure virality by counting shares, likes, comments, and impressions. Number of followers, network centrality, and community membership can reveal how information travels through social networks and the influence of specific users or groups (Huynh, et al., 2022). Temporal data, including timestamps of posting and engagement events, track when and how quickly content gains traction.

Source credibility is crucial in determining the trustworthiness of information. Data on account type (influencer, expert, news outlet, regular user) and verification status help assess the reliability of those disseminating content. Algorithmic signals, such as whether a post is boosted, labeled, or suppressed by platform algorithms, inform how the platform itself influences the spread of information and can significantly affect engagement.

Exposure to prebunking or media literacy interventions should also be considered. Data on whether users have interacted with educational content about misinformation can shed light on the effectiveness of such measures in promoting the dissemination of true information over falsehoods.

Data for this research can be sourced from public social media datasets (e.g., Twitter/X API, Facebook CrowdTangle, Reddit Pushshift, YouTube Data API), fact-checking organizations (Full Fact, Snopes, PolitiFact, Community Notes), and academic resources like MIT's "False News" dataset, PHEME, and FakeNewsNet. Surveys and experiments can provide insights into user sharing habits and responses to prebunking. Platform transparency reports may offer data on algorithms and engagement. For unavailable data, custom methods such as permitted web scraping or collaborating with platforms for anonymized data can be used.

Data Preparation Plan

For robust analysis, datasets from multiple platforms will be integrated, variables and timestamps standardized (UNIX), and engagement metrics normalized. Fact-checking databases will label posts as true or false, while natural language processing (NLP) models

will assess sentiment and content novelty for credibility and virality insights (Hasan, Maisha, & Arifuzzaman, 2019).

Missing engagement metrics will be imputed when possible; posts with essential missing data will be excluded. The dataset will be cleaned of duplicates, bots, and spam, and filtered by relevant periods and topics. Data sources, including behavioral and survey data, will be merged to offer a comprehensive perspective and linked to fact-checking interventions like Community Notes.

All data handling will follow ethical standards, ensure user anonymity, and comply with platform policies and privacy laws. We will expound on this later.

Data Analysis Plan

Summary statistics will be calculated to determine the averages, medians, and distributions of engagement metrics, with a focus on distinguishing between true and false posts. Following the example of Sangiorgio et al. (2025), we can illustrate virality trends using time series plots, and network graphs to map patterns of information spread, while sentiment heatmaps and bar charts will compare engagement across content types.

Correlation analyses will examine relationships between emotional tone and virality, as well as between source credibility and engagement metrics (Trunfio & Rossi, 2021). Comparative analyses will look at differences between posts that receive algorithmic boosts or fact-check labels and those that do not. The effect of influencer amplification will be contrasted with grassroots sharing (Huynh, et al., 2022). Topic modeling will help identify recurring themes in viral content, distinguishing between true and false information.

Modeling Approach

Here, we define the analytical frameworks and statistical techniques that will be employed to distinguish between factual and misleading posts, predict their virality, and uncover the underlying dynamics of information dissemination.

Models

We will use classification models including logistic regression, decision trees, support vector machines, and deep learning models to distinguish between true and false (or misleading) posts based on features such as content, sentiment, source credibility, and engagement metrics. Classification is necessary for automated detection and labeling of misinformation versus factual content, a foundational step in combating disinformation (Ashiq, et al., 2023).

Doe et al. (2023) demonstrated that regression and survival analysis provide insight into what drives virality and how quickly information spreads, which will allow us to develop targeted interventions. Regression models will predict the degree of virality (e.g., number of shares, likes, or comments) for posts labeled as true or false, using features like emotional tone, novelty, and network centrality. Survival analysis will model the time until a post is reshared, which will reveal the speed of information spread.

Clustering will identify natural groupings of posts or users based on engagement patterns, sentiment, or network structure, which can reveal underlying dynamics of viral spread. Network Analysis can map and analyze the diffusion pathways of true and false information, identifying influential nodes and structural differences in spread. Clustering

and network analysis help uncover patterns and influential actors, which are critical for designing strategies to amplify true information (Kumar & Shah, 2018).

Model Evaluation

To enhance model usefulness and avoid overfitting, models will be evaluated based on the following criteria:

- Accuracy, the proportion of correctly classified posts (true/false).
- Precision, the proportion of posts correctly classified as true.
- Recall, the proportion of actual true posts that are correctly identified.
- F1 Score, the harmonic mean of precision and recall (Ashiq, et al., 2023).
- AUC-ROC, which measures the ability of the model to distinguish between classes across all thresholds.
- Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which assess prediction accuracy for engagement metrics in regression models.
- Concordance index (C-index), to evaluate time-to-event predictions (Doe, et al., 2023).

Best Practices

To ensure model generalizability, we will implement cross-validation procedures throughout the development process. We can avoid class imbalance by applying resampling techniques or weighted loss functions. Evaluating model performance across diverse datasets will help identify potential weaknesses and validate the effectiveness of the model in various real-world scenarios (Chung & Chen, 2025).

Assumptions, Challenges, and Ethics

Assumptions

Our models make several key assumptions regarding data labeling and metric validity. We presume that the labels distinguishing true from false information are both accurate and consistently applied across various datasets, minimizing the risk of introducing bias or error during model training and evaluation. We treat engagement metrics as reliable proxies for measuring the virality of content. We also assume that sentiment and novelty scores effectively capture the emotional and informational impact of posts, reflecting how users may perceive and interact with the content. Inconsistencies or inaccuracies in any of these areas could significantly affect overall model performance and interpretation.

Challenges

A study of this magnitude and complexity will introduce some challenges and face several limitations:

- Many misinformation datasets suffer from "spurious correlations that could lead to predictions based on invalid… signals" (Thibault, et al., 2025).
- Models trained on one platform or dataset may not perform well on others due to differences in user behavior and platform algorithms (Chung & Chen, 2025).
- The speed and pattern of information spread can change rapidly due to "external events" or "platform interventions" (Doe, et al., 2023).
- Viral spread is influenced by network structure, influential users, and algorithmic amplification, which "are difficult to fully capture" (Vosoughi, Roy, & Aral, 2018).

- Automated detection and suppression of content can lead to unintended consequences for freedom of expression, such as "privacy violations, and AI-driven bias" (Skandali, 2025).

Ethics

We will respect user privacy by ensuring all data collection and analysis comply with privacy laws and platform policies. PII will be anonymized to protect individual identities.

Model limitations and decision criteria will be transparently communicated to a diverse group of stakeholders. We will avoid opaque "black box" models where possible (Skandali, 2025). Because models may inadvertently reinforce existing biases in data or platform algorithms, they will be regularly audited for bias against particular groups or viewpoints. New forms of misinformation and platform changes will also require continuous model adaptation.

When developing intervention strategies, we will consider the broader consequences such as risk of eroding trust or amplifying polarization. Overzealous filtering of content can suppress legitimate debate and dissent. Interventions may shift misinformation to less visible channels or increase distrust in platforms.

Conclusion

The challenge of making truth as compelling and widely shared as false information on social media requires a nuanced approach that draws on both complex data modeling and a deep understanding of human behavior. If research successfully identifies strategies that increase the appeal and dissemination of accurate information, the implications are profound. Such advances could not only curb the spread of misinformation but also foster

more informed, engaged, and trusting online communities. Ultimately, promoting the truth in digital spaces has the potential to strengthen democratic discourse, reduce polarization, and build resilience against manipulation and deception. The success of these efforts would represent a critical step toward a healthier and more trustworthy information ecosystem.

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